

3-Level Geometric Modeler: CSG, Subdivision Trees and Boundary Representation

KAI CAO, ZHONGDI LUO

New York University
kcao@cs.nyu.edu

Abstract

The problem of computing geometry modeling in $\mathbb{R}^3$ is fundamental to computer graphics, path finding, motion planning and numerous branches of computer science. Given a polyhedron P defined by constructive solid geometry (CSG) model, we raise a new idea to present the boundary features (vertices, edges and faces) of P using subdivision method; it has the following steps: 1) define P in CSG model, 2) subdivide a given box B_0 for P and construct the adequate subdivision tree T , 3) represent the boundary features based on the subdivision tree T . We could also compute the boolean operations of 2 subdivision trees. Since the great generality and scalability of subdivision method, this idea can also be expanded to support more complex shape with higher order surfaces in $\mathbb{R}^3$.

I. INTRODUCTION

In this paper, we focus on the geometric modeling of solids (especially polyhedra) in $\mathbb{R}^3$. The traditional definition of a polyhedron in $\mathbb{R}^3$ is the boolean union of half spaces (CSG). The faces of a polyhedron P are the plains which bounds the half spaces. Meanwhile the edges are the intersection of two unparallel plains and the vertices are the intersection of three. Faces, edges and vertices constitute the features of a polyhedron. By CSG definition, these features are not obvious. To present them more efficiently and make them more operable, we keep subdividing a box B_0 containing P until the boundary features are kept in certain subboxes. Then we can compute the features by manipulating the subdivision tree instead.

II. REGULARIZED SET

A set $S \subset \mathbb{R}^3$ is a **regularized set** if $S = \overline{\text{int}(S)}$.

A **solid** is defined as a regularized subset of $\mathbb{R}^3$. Obviously a polyhedron P is a finite

solid.

Suppose A and B are regularized sets, we define the boolean operations:

- $A \cup B = \overline{\text{int}(A \cup B)}$
- $A \cap B = \overline{\text{int}(A \cap B)}$
- $A - B = \overline{\text{int}(A - B)}$
- $A^c B = \overline{\text{int}(A^c)}$

III. CSG PRIMITIVES

A **constructive solid geometry** (CSG) model describes a polyhedron as regularized set operation on primitives which are the simplest solid objects used for the representation.

Primitives in CSG models are all semi-algebraic sets as follows:

1. **Half Space**: $H = \{(x, y, z) | a * x + b * y + c * z + d \geq 0\}$.
2. **Sphere**: $S = \{(x, y, z) | (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 - r^2 \leq 0\}$.

$$3. \text{ Cylinder : } C = \{(x, y, z) | A = -x \sin(\theta) + y \cos(\theta) \cos(\phi) + z \cos(\theta) \sin(\phi), \\ B = -y \sin(\phi) + z \cos(\phi), A^2 + B^2 \leq R^2\}.$$

$$4. \text{ Torus : } T = \{(x, y, z) | (R - \sqrt{x^2 + y^2})^2 + z^2 \leq r^2\}.$$

A polyhedron is limited by the intersection of half spaces.

IV. SUBDIVISION TREE

4.1 Box

B is a **box** if $B \subset \mathbb{R}^3$, $B = \{(x, y, z) | x_1 \leq x \leq x_2, y_1 \leq y \leq y_2, z_1 \leq z \leq z_2\}$, or simplified as $B = ([x_1, x_2], [y_1, y_2], [z_1, z_2])$, which is a regularized subset of $\mathbb{R}^3$.

4.2 Subdivision on box

Suppose $B = ([x_1, x_2], [y_1, y_2], [z_1, z_2])$, let $\bar{x} = \frac{x_1+x_2}{2}$, $\bar{y} = \frac{y_1+y_2}{2}$, $\bar{z} = \frac{z_1+z_2}{2}$. The subdivision on B is to subdivide B into eight subboxes by splitting at $x = \bar{x}$, $y = \bar{y}$ and $z = \bar{z}$.

4.3 Features

For a polyhedron $P \subset \mathbb{R}^3$, we use set V , E and F to represent its vertices, edges and faces respectively. The **features** $\Phi(P) = (V, E, F)$.

Then we define the features of a box B corresponding to polyhedron P as follows:

- $\Phi_0(B, P) = B \cap V$
- $\Phi_1(B, P) = \{e \mid e \in E, e \cap B \neq \emptyset\}$
- $\Phi_2(B, P) = \{f \mid f \in F, f \cap B \neq \emptyset\}$

$$\Phi(B, P) = (\Phi_0(B, P), \Phi_1(B, P), \Phi_2(B, P))$$

4.4 Elementary Box

We define a box B to be **elementary box** for a polyhedron P if one of the following conditions hold:

$$1. B \cap P = \emptyset.$$

Call B an **Empty Box** for P .

$$2. B \subseteq P.$$

Call B a **Full Box** for P .

$$3. \text{ Exists only one } v \text{ st. } \Phi_0(B, P) = \{v\} \text{ and } \forall e \in \Phi_1(B, P) \text{ } e \text{ is bounded by } v \text{ and } \forall f \in \Phi_2(B, P) \text{ } f \text{ is bounded by } v.$$

Call B a **Vertex Box** for P .

$$4. V \cap B = \emptyset, \text{ exists only one } e \text{ st. } \Phi_1(B, P) = \{e\} \text{ and } \forall f \in \Phi_2(B, P) \text{ } f \text{ is bounded by } v.$$

Call B a **Edge Box** for P .

$$5. V \cap B = \emptyset, E \cap B = \emptyset, \text{ exists only one } f \text{ st. } \Phi_2(B, P) = \{f\}.$$

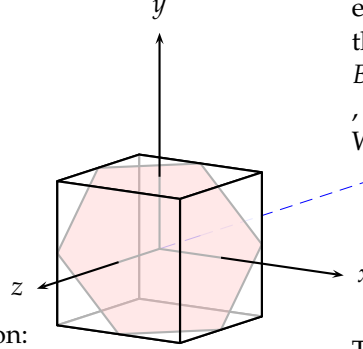
Call B a **Face Box** for P .

4.5 Definition

1. A **subdivision tree** is a tree where each interior node has degree 8 (Octree) and each node u is associated with a box $u.box$ st. if $u_0, \dots, u_7$ are the children of u . Then $\{u.Box : i = 0..7\}$ is the split of $u.box$.
2. An **adequate** subdivision tree of a box B_0 for a polyhedron P is a subdivision tree rooted at B_0 , in which all leaves are associated with elementary boxes for P .

4.6 Classification of elementary boxes

1. Face box B for polyhedron P , the face $f \cap B \neq \emptyset$



(a) $f \cap P$ is a hexagon:

$T_2 = AST(B_s, P_2)$ rooted at the same box B_s :

$T = M(BO, T_1, T_2)$ in which BO is a Boolean operation and T_1 and T_2 are adequate subdivision trees. The result of the function M is an AS-Tree T .

BO are Boolean operations as union ($\cup$), intersection ($\cap$) and difference ($-$).

We also define:

- $T_1 \cup T_2 = M(\cup, T_1, T_2)$
- $T_1 \cap T_2 = M(\cap, T_1, T_2)$
- $T_1 - T_2 = M(-, T_1, T_2)$

The result $T = AST(B_s, P_1 \text{ BO } P_2)$.

V. BOOLEAN OPERATIONS ON SUBDIVISION TREES

To compute the features of the union, intersection or difference of two polyhedra. We could compute the boolean operations on their subdivision trees.

Give box B_0 and polyhedron P_1, P_2 . We have two adequate subdivision trees T_1 for P_1 , T_2 for P_2 .

Before we do the boolean operation on T_1, T_2 , we need to make them congruent, which means $\forall$ node n_1 in $T_1, \exists n_2$ in T_2 st. $n_1.\text{box} = n_2.\text{box}, \forall$ node n_2 in $T_2, \exists n_1$ in T_1 st. $n_2.\text{box} = n_1.\text{box}$. We name this process **further subdivision**.

1. Union

Boolean operations on two adequate subdivision trees $T_1 = AST(B_s, P_1)$ and

VI. BOOLEAN OPERATION TWO EST-NODES $n(B_e, P_1)$ AND $n(B_e, P_2)$

6.1 Union

1. If B_e is an empty box for either P_1 or P_2 (let us say P_1):

$$n(B_e, P_1) \cup n(B_e, P_2) = n(B_e, P_1 \cup P_2) = n(B_e, P_2)$$
2. If B_e is a full box for either P_1 or P_2 (let us say P_1):

$$n(B_e, P_1) \cup n(B_e, P_2) = n(B_e, P_1 \cup P_2) = n(B_e, P_1)$$
3. If B_e is a face box for both P_1 and P_2 :

(a)

(b)

(c)